AI-Powered Research Gap Identifier & Trend Analyzer

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Selecting a suitable supervisor is a critical first step for students and early-stage researchers, but the process is often slow, informal, and unfair. In many universities, students and early-stage researchers still rely on informal methods such as personal contacts, trial-and-error emailing, or limited profile browsing. Even when automated supervisor recommendation systems are used, most existing approaches mainly focus on topic similarity or static scoring criteria. This creates a major practical issue: a supervisor may match the topic well but may be overloaded, unavailable, or already committed to multiple projects. As a result, recommendations can be unrealistic, supervision distribution becomes unfair, and research progress is delayed.

This research proposes a Recommendation & Collaboration Network module that matches a researcher's research details with suitable supervisors by combining topic relevance with academic relationship knowledge. The module first structures supervisor profiles and research idea metadata, then builds an academic collaboration network that represents supervision history, co-supervision, research groups, and collaboration links. A key novelty is using a graph neural network (GNN) as the main recommendation engine so the system learns from both profile content and network structure instead of relying only on direct keyword overlap. The model is designed to handle a continuously growing environment where new supervisors register and new research topics arrive without hardcoded updates. In addition, supervisor workload is treated as a core feasibility gap: workload signals are incorporated into ranking so the system avoids repeatedly recommending supervisors who are already heavily loaded.

The expected outcome is a recommendation workflow that produces more accurate and feasible supervisor suggestions, improves fairness by balancing recommendation exposure across staff, and reduces the time researchers spend searching for guidance. Social impact includes improved access to mentorship and faster research initiation, while commercial impact includes a deployable university-facing module that supports supervision management and collaboration building.

Keywords: supervisor recommendation, collaboration network, graph neural network, workload-aware ranking, research management

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My appreciation is also extended to my colleagues and peers for their helpful discussions, suggestions, and feedback during the preparation of this proposal. Their support and collaborative input played an important role in improving the clarity and practicality of the proposed solution.

This research proposal represents an important step toward developing a Recommendation and Collaboration Network that supports more accurate, feasible, and workload-aware supervisor matching, contributing to a more efficient and balanced research supervision process.

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LIST OF ABBREVIATIONS

GNN	Graph Neural Network
CoEAI	Centre of Excellence for AI

Table 1: Table of Abbreviations

1 INTRODUCTION

1.1 Background and Context

1.1.1 Describe the real-world problem

Finding the right research supervisor is currently an inefficient and often informal process at most universities. Students typically rely on outdated departmental webpages, requesting by meeting physically, or speculative emails, leading to a "wait and see" approach that lacks transparency. Research shows that selecting a supervisor based on limited information can result in poor academic outcomes, especially when choices are made without analyzing a supervisor's full publication record or past history. As demand for supervision grows, simply matching a student to a topic is no longer enough. Even a perfect topical match can fail if the supervisor's workload is overwhelmed or lacks the specific collaboration network needed to support the student's long-term goals. This creates a systemic problem where a lack of data on workload and networking leads to mismatched pairings and uneven research support.

1.1.2 Who is affected and why it is important?

Stakeholder Group	Nature of Impact	Severity
University Students / Early-stage Researchers	Delays in finding suitable supervisors, repeated trial-and-error outreach, mismatched supervision that slows research progress and reduces guidance quality	High
Academic Supervisors	Uneven supervision distribution (popular supervisors overloaded), reduced feedback quality and availability, increased administrative coordination for co-supervision and projects	High
Department / Faculty Research Coordinators	Difficulty ensuring fair allocation, limited visibility into workload and expertise distribution, manual effort required to manage supervision capacity and resolve conflicts	High
University / Research Institution	Lower overall research efficiency, delayed completions, reduced research output quality, governance challenges in transparency and accountability of supervision allocation	Medium
Research Groups / Labs	Missed collaboration opportunities due to poor visibility of networks, weak alignment between student topics and active research clusters, fragmented project planning	Medium

Society / Knowledge Economy	Slower development of research talent, reduced innovation pipeline, weaker alignment between academic research and national development priorities	Medium

Table 2: Stakeholder Group

1.1.3 Statistical evidence (local/global)

Local access pressure : National-level statistics show that in 2021, only 25.39% of eligible students had a chance to enter the national university system. This indicates high competition for academic resources and supervision capacity, increasing the importance of efficient, transparent allocation mechanisms.

Global completion pressures : Large-scale evidence shows that PhD completion varies by field and is not universally high; one cited analysis reports 10-year completion rates of 64% down to 49% . While completion depends on many factors, supervisory support and feasibility are repeatedly identified as critical components of the doctoral experience.

1.1.4 Analysis of Existing Solutions and their limitations

Content-based matching - TF-IDF / cosine similarity

Many systems match a researcher's title or abstract or keywords with supervisor profiles or publication text using TF-IDF and cosine similarity. This is simple, explainable, and works even without historical interaction data. However, it mainly depends on keyword overlap, so it can miss semantic similarity and usually does not consider feasibility factors such as supervisor workload and availability.

Hybrid filtering - content-based and collaborative signals

Hybrid approaches combine text relevance with interaction history to improve recommendation robustness. These systems often outperform single-method approaches, but they commonly use fixed linear weights, struggle with cold-start situations for example new supervisors/new topics, and require interaction data that may be sparse. Privacy and fairness issues are also frequently identified as concerns for real deployment.

1.1.5 Ideas for the proposed solution and its parts

The proposed solution is a Recommendation & Collaboration Network module that matches a researcher's research details (title, abstract, keywords) with suitable supervisors and improves the recommendation quality using collaboration-network intelligence and workload feasibility.

Profile and metadata structuring - The system will first collect and store structured supervisor/researcher profiles and research idea metadata in a database. This ensures that recommendations are based on consistent and updateable information rather than manual searching.

Profile representation for matching. Each supervisor and research idea will be converted into a machine-readable representation. Supervisor profiles will be represented using expertise keywords and research text, while the researcher's idea will be represented using the submitted title/abstract/keywords. These representations enable systematic similarity computation and graph-based learning.

Supervisor matching and ranking. The system will generate an initial shortlist of supervisors by estimating topic relevance between the research idea representation and supervisor profile representations. This provides a clear, explainable baseline ranking.

Collaboration network construction. A collaboration network will be built using available relationships such as co-authorship, co-supervision, research group membership, and topic associations. This network captures "who works with whom" and helps identify supervisors who are active and connected in the relevant research area.

Graph-based improvement and feasibility handling. Recommendations will be refined using network signals through graph scoring and a learning-based approach. A GNN will be used to learn representations from the dynamic graph and improve ranking quality, while workload indicators will be incorporated to penalize or filter overloaded supervisors. The final output will provide top-K supervisors with simple justification.

1.1.6 Relevant SDG alignment

SDG 4 – Quality Education:

By helping students and early-stage researchers quickly identify suitable supervisors and research groups, the component improves access to guidance and reduces delays in starting research. Better supervisor–topic fit and feasible supervision allocation support stronger learning outcomes, improved research quality, and more effective academic development.

SDG 9 – Industry, Innovation and Infrastructure:

The component contributes to building digital research infrastructure by introducing a structured, data-driven recommendation pipeline .This supports innovation in research management and enables universities to operate more efficiently using modern AI and graph-based technologies.

1.1.7 Research cluster alignment

The Recommendation & Collaboration Network component aligns directly with the CoEAI research cluster by applying AI-driven methods to solve a real institutional problem in research management—matching researchers to suitable supervisors and identifying collaboration opportunities.

This component uses key CoEAI-relevant themes such as intelligent recommendation systems**,** graph-based modeling**,** and representation learning to capture both research-topic relevance and academic relationship structure. In addition, the workload-aware design supports responsible and practical AI adoption within academic environments by focusing on feasibility and fairness, which are important for scalable institutional deployment.

1.2 Literature Review

Paper (Year)	**Technologies**	**Tools**	**Stack**	**Methodology**	**Problem & results**	**Limitations**
Criteria-based Ph.D. Supervisor Recommendation-2018 [1]	content-based, AHP, fuzzy	Java, Elasticsearch, MAG, CORE	Java + Elasticsearch + MAG + CORE + content-based + AHP + fuzzy	supervisor recommendation based on multiple criteria. Student preferences and supervisor attributes are collected as criteria. Topic fit is estimated using text similarity/ranking ideas. The final recommendation is a ranked shortlist based on combined scores rather than only text matching.	Problem: Support PhD students in selecting supervisors using structured criteria. Result: Improves decision support compared to informal/manual selection.	Weights can be subjective; often limited evaluation scope; typically missing workload/ availability constraints and collaboration networks.

Deep Learning Recommendation Model for personalization and recommendation systems-2019 [2]	deep learning, neural, embedding, precision, recall	Python, NumPy, Angular, MAG	Python + NumPy + Angular + MAG + deep learning + neural + embedding	The model learns vector representations (embeddings) for users and items from data. A neural network learns complex patterns beyond simple similarity matching. After training, it scores candidate items for a user and ranks them. This usually works well when there is lots of data.	Problem: improving personalization/recommendation using deep learning. results: improves ranking/selection compared to simple baselines	Needs large datasets and compute; harder to explain; may amplify bias if not controlled.
Multi-Criteria Decision Support System for Ph.D. Supervisor Selection A Hybrid Approach-2019 [3]	TF-IDF, BM25, Okapi, hybrid, AHP, fuzzy, recall	Elasticsearch, MAG, CORE	Elasticsearch + MAG + CORE + TF-IDF + BM25 + Okapi + hybrid	defines criteria for supervisor choice (topic fit + other factors). Collects student preferences and supervisor attributes. Computes weighted scores and ranks supervisors. Designed to support decision-making rather than only text matching.	Problem: recommend/allocate suitable supervisors for students. results: improves ranking/selection compared to simple baselines	Criteria weights can be subjective; often not adaptive; collaboration networks and workload constraints may be weak.
Supervisor Recommendation Tool for Computer ScienceProj	TF-IDF, content-based	Python, NLTK, CORE, WordNet, RAKE, Porter	Python + NLTK + CORE + WordNet + TF-IDF + content-based	The tool extracts key terms from the student's project idea. It builds supervisor expertise profiles from research	Problem: Speed up CS project supervisor matching and reduce random selection. Result: Demonstrates	Mostly word-based matching; weak on semantics, capacity/workload constraints, and collaboration-

ects-2019 [4]				interests/ publications. It computes similarity between student keywords/text and supervisor profiles. Supervisors are ranked and the top matches are recommended.	improved matching over manual/naive selection in their setting.	network evidence.
Multi-Criteria Decision Support System for Recommendation of Ph.D. Supervisor-2020 [5]	TF-IDF, cosine similarity, collaborative filtering, content-based, hybrid, AHP	Java, NetBeans, Google Scholar, Web of Science, MAG, CORE, RAKE	Java + NetBeans + Google Scholar + Web of Science + TF-IDF + cosine similarity + collaborative filtering	The approach lists criteria for supervisor selection and compares them using AHP pairwise comparisons. AHP produces weights for each criterion. Each supervisor is scored using those weights and ranked. The output is a supervisor shortlist based on multi-criteria decision making.	Problem: Help students choose supervisors when many criteria matter. Result: Produces a structured ranking that supports decision-making.	Weights can be subjective; may require manual inputs; often missing collaboration graphs and constraint-aware filtering.
Research Supervisor Recommendation System-2020 [6]	TF-IDF, Okapi, cosine similarity, collaborative filtering, content-based, hybrid, fuzzy, neural	PHP, MySQL, XAMPP, MAG, CORE	PHP + MySQL + XAMPP + MAG + TF-IDF + Okapi + cosine similarity	The system collects student topic/proposal text and supervisor expertise text. It cleans the text (remove noise, tokenize, remove stop words). It converts text to vectors (TF-IDF) and computes cosine similarity. Supervisors are ranked by similarity and the top matches are recommended.	Problem: Recommend supervisors by matching student topic to supervisor expertise. Result: Demonstrates workable recommendations and reports evaluation results within their dataset.	Word-based matching; weak on synonyms/semantics; often missing constraints like workload/availability and collaboration-network evidence.
Thesis Supervisor Recommendation with Representative-2020 [7]	cosine similarity, collaborative filtering, content-based, hybrid, precision, recall	PHP, MAG, CORE	PHP + MAG + CORE + cosine similarity + collaborative filtering + content-based + hybrid	Representative-profile method: builds a representative text profile for each supervisor from past supervised work. Uses the student's short title/abstract as the query. Computes	Problem: Improve recommendations when student input is short or vague. Result: Shows better ranking than using only the raw short query in their evaluation.	Representative profiles can become outdated; still text-similarity based; does not automatically include workload/capacity constraints.

				similarity between the query and each representative profile. Ranks supervisors and recommends the top matches.		
Recommender System for an Academic Supervisor with a Matrix Normalization Approach (2021) [8]	hybrid, fuzzy, context-aware	Requests, Scopus, CORE	Requests + Scopus + CORE + hybrid + fuzzy + context-aware	The approach collects supervisor indicators from bibliographic data. It builds a scoring matrix across supervisors and normalizes values so they can be compared fairly. Supervisors are ranked using the normalized scores. This is useful when structured data is available and consistent.	Problem: Recommend supervisors using structured scholarly indicators rather than only topic text. Result: Produces ranked supervisors aligned with bibliographic signals.	Depends on database access; can favor very prolific supervisors; topic-fit and collaboration context may be weak if not added.
Using Academic Genealogy for Recommending Supervisors (2021) [9]	TF-IDF, cosine similarity, collaborative filtering, content-based, hybrid, fuzzy, context-aware, genealogy, Nearest Centroid, precision	Python, Elasticsearch, NLTK, scikit-learn, MAG, CORE	Python + Elasticsearch + NLTK + scikit-learn + TF-IDF + cosine similarity + collaborative filtering	builds an academic genealogy network from thesis/dissertation metadata. It uses thesis titles/abstracts and the genealogy links as evidence. Given a new query, it scores and ranks supervisors using both topic and network signals. This helps recommend supervisors based on real advising relationships.	Problem: Recommend supervisors using advising-relationship structure, not only publication keywords. Result: Demonstrates effective ranking on their dataset using IR-style metrics.	Genealogy data can be incomplete or outdated; does not automatically handle supervisor availability/workload; needs good name disambiguation.
A Survey of Recommendation Systems: Recommendation Models, Techniques, and Application Fields (2022) [10]	TF-IDF, cosine similarity, collaborative filtering, content-based, hybrid, AHP, fuzzy, deep learning, neural, context-aware	React, Google Scholar, MAG, CORE	React + Google Scholar + MAG + CORE + TF-IDF + cosine similarity + collaborative filtering	reviews the main types of recommender systems and explains how they work at a high level. It compares different models (content-based, collaborative, hybrid, etc.) and where they are used. It also summarizes common evaluation methods and recurring problems. You mainly	Problem: Provide an overview of recommender system research. Result: A clear taxonomy of models + common issues (cold-start, sparsity, bias, scalability).	no end-to-end solution for university constraints.

				use it to justify your method choice and to show the known challenges.		
Recommendation System to Propose Final Project Supervisor Using Cosine Similarity (2022) [11]	TF-IDF, cosine similarity, collaborative filtering, content-based, AHP, neural	Python, Java, Django, React, Selenium, BeautifulSoup, Requests, Bootstrap, Google Scholar, MAG	Python + Java + Django + React + TF-IDF + cosine similarity + collaborative filtering	The system collects supervisor profile/publication text and student project descriptions. It preprocesses text and converts it into vector form. Cosine similarity is used to score how close each supervisor is to the student topic. Supervisors are ranked and shown through a web interface.	Problem: Automate supervisor recommendations for final-year projects. Result: Provides a working web system and shows evaluation/validation in their context.	Cosine similarity is shallow semantics; data collection (especially scraping) can be noisy; often missing fairness and workload constraints.
Exploring Research Supervisor Recommendation: Student-Centric or Supervisor-Centric (2023) [12]	TF-IDF, collaborative filtering, content-based, hybrid, matrix normalization	MAG, CORE	MAG + CORE + TF-IDF + collaborative filtering + content-based + hybrid + matrix normalization	It explains how the target user changes the ranking objective and criteria. It discusses trade-offs such as fairness, workload balance, and usability. It is useful to position your system and justify your design choices.	Problem: Clarify how supervisor recommenders should be designed depending on who the main user is. Result: Provides guidance and highlights trade-offs rather than building a full system.	No end-to-end implemented recommender; limited experimental validation.
Research Supervisor Recommendation System Using a Hybrid Filtering Approach (2025) [13]	TF-IDF, cosine similarity, collaborative filtering, content-based, hybrid, embedding, precision, recall, AUC, Hit@	Python, Pandas, NumPy, Matplotlib, CORE	Python + Pandas + NumPy + Matplotlib + TF-IDF + cosine similarity + collaborative filtering	The system computes a content-based score from topic text (student ↔ supervisor). It also computes a collaborative score from interaction history (who selected/worked with whom). Both scores are combined (often with weights) to rank supervisors. It is evaluated using ranking metrics such as Hit@K and AUC.	Problem: Improve supervisor recommendation beyond content-only matching. Result: Shows better ranking performance than single-method baselines in their experiments.	Mostly word-based matching; weak on semantics, capacity/workload constraints, and collaboration-network evidence.
Two-Way Thesis Supervisor	TF-IDF, hybrid, AHP, MapReduce,	Python, Java, PHP, Flask,	Python + Java + PHP + Flask + TF-IDF +	The system represents supervisors and students using	Problem: Efficient multi-criteria matching for	More complex than needed for small datasets; requires

Recommendation System Using MapReduce K-Skyband View Queries (2025) [14]	K-Skyband, Skyband, skyline, deep learning	MySQL, Jupyter, Visual Studio Code, MAG, CORE	hybrid + AHP	multiple attributes (dimensions). K-skyband selects candidates that are strong across dimensions (not dominated). MapReduce-style processing is used for efficiency on larger data. The final output is a shortlist of strong matches for both sides (two-way matching).	supervisor recommendation at scale. Result: Improves efficiency and supports multi-attribute selection compared to naive ranking.	clean feature engineering; topic semantics may be weak unless combined with NLP/IR.

Table 3: Literature Review

1.3 Research Gap

Existing supervisor recommendation research provides useful methods for topic matching**,** hybrid ranking, and multi-criteria decision support, but it largely fails to produce recommendations that are both topically relevant and feasible under real supervision capacity. In particular, supervisor workload is often ignored, represented as a simple proxy, or treated as a manual constraint rather than being modeled using the academic network structure. This means current systems can recommend supervisors who match the research topic well but are already overloaded, reducing practicality and fairness.

Topic-only similarity (TF-IDF/cosine): approaches are widely used because they are simple and explainable. However, they are strongly dependent on lexical overlap and often ignore feasibility constraints; topic-only systems also struggle when profiles are short, outdated, or expressed using different vocabulary. [6]

Hybrid filtering: improves robustness by combining content-based relevance with collaborative/interaction signals. Yet, one recent hybrid supervisor recommendation study explicitly reports limitations such as reliance on a simple linear weighted combination, shallow text representation compared to stronger semantic modeling, cold-start limitations for new supervisors/topics, and the need to address privacy and fairness for real deployment. The same paper also highlights that results are often validated within a limited institutional dataset, restricting generalization. [13]

Relationship-aware: Academic genealogy-based recommendation shows that advising relationships and social context can improve matching; however, the same line of work still uses classical vector-space representations and explicitly prioritizes precision while disregarding scalability. It also reinforces that supervisor choice requires analyzing distributed repositories and that social network relationships matter—yet it does not operationalize current workload feasibility as a ranking constraint. [8]

Multi-criteria decision support: systems introduce additional criteria beyond topic fit, but often require subjective weighting or user-provided judgments that are difficult for students to estimate

accurately. Prior work also notes that collaboration networks matter, yet their influence is often not fully assessed from a student-centric perspective and is not used to realistically quantify supervision feasibility. [3]

This research introduces a novelty that directly addresses these limitations: it proposes combining the strongest ideas from prior methods (content-based matching, hybrid filtering, multi-criteria decision support) with a GNN. Instead of treating workload as a static number or a manual rule, workload is modeled as a learnable outcome influenced by supervision relationships and collaboration structure. This directly responds to gaps acknowledged in the literature, including fixed weighting limitations, cold-start challenges, and fairness concerns.

Therefore, the research gap is the absence of explicitly incorporating workload feasibility and addressing cold-start and fairness concerns reported in existing supervisor recommender studies.

2 OBJECTIVE

2.1 Main Objective

To develop and evaluate a workload-aware supervisor recommendation module that combines content-based matching, hybrid and multi-criteria scoring, and a GNN-based workload estimation model to produce Top-K supervisor recommendations that are both topically relevant and feasible under workload constraints, with performance validated using standard ranking and workload-feasibility metrics.

2.2 Specific Objectives

Profile & metadata structuring:
Collect and store structured supervisor profiles and researcher/project idea metadata (title, abstract, keywords) in a database to support automated matching.

Representation generation:
Generate machine-readable representations for supervisors and research ideas (text/profile vectors and structured features) that enable baseline matching and graph-based learning.

Baseline recommendation implementation:
Implement baseline supervisor recommendation methods from literature and generate Top-K supervisor lists for given research ideas.

Collaboration network construction:
Build a collaboration network graph using available relationships and compute network-based features/scores to enhance recommendation quality.

GNN-based workload estimation:
Design a measurable workload proxy/label and train a GNN model to predict supervisor workload from graph structure and node features.

3 METHODOLOGY

3.1 Tools and Platforms

The component uses PostgreSQL/MySQL as the primary data store to keep supervisor profiles, research idea metadata, and collaboration/supervision relationships in a structured form. This is essential for maintaining consistency, supporting updates, and enabling integration with other system modules. Python is used as the backend language because it supports data processing, graph construction, and direct integration with modern graph learning frameworks.

For the GNN implementation, the component uses PyTorch (PyG). PyTorch provides a stable foundation for training neural models, while PyG provides specialized graph learning support, including heterogeneous graph handling and inductive GNN architectures suitable for dynamic networks. A lightweight API framework such as FastAPI/Flask can be used to expose endpoints for supervisor recommendations and to integrate this component into the overall system. For experimentation and reporting, standard development environments such as notebooks and code editors can be used to train models, log results, and produce evaluation outputs.

A GNN is used because supervisor recommendation depends on relationships, not only on keywords. A GNN learns directly from this collaboration graph and captures multi-hop connections, whereas if/else rules require manual thresholds and a standard neural network would need the graph to be flattened into handcrafted features. This makes the approach better suited for producing realistic recommendations in a dynamic environment where new supervisors and topics continuously appear. A more suitable option is to use an inductive heterogeneous GNN such as GraphSAGE on the graph to learn node embeddings. GraphSAGE is suitable because it supports inductive learning on growing graphs.

3.2 Data sources

- Supervisor profiles: interests or keywords, department, group membership, as another resource publication metadata.
- Supervision history: past project–supervisor links.
- Collaboration links: co-authorship/co-supervision/group links
- Workload indicators: active supervisee counts and co-supervision commitments.

3.3 Ethics and privacy controls

- **Data minimization:** store only fields needed for matching and evaluation.
- **Access control:** restrict who can view workload and recommendation logs.
- **Anonymization for experiments:** during training and evaluation, replace names with IDs; keep a separate secure mapping if needed.
- **Bias and fairness monitoring:** check whether recommendations systematically favor a small subset of supervisors; incorporate workload constraints to reduce unfair concentration.
- **Consent/approval:** if real institutional records are used, follow department/university approval procedures and document the handling policy.

3.4 Validation Strategy and Performance Metrics

The proposed Recommendation & Collaboration Network component will be evaluated to measure its effectiveness in generating relevant and feasible supervisor recommendations for a given research idea.

Evaluation metrics may include:

- **Recommendation Accuracy** – How well the recommended supervisors match the research topic and keywords
- **Precision and Recall** – Quality of the Top-K recommended supervisors
- **Ranking Performance** – Effectiveness of the ranking of supervisors.
- **Workload Feasibility Evaluation** – Ability to reduce recommendations of overloaded/unavailable supervisors

The results of the proposed GNN-based approach will be compared with traditional supervisor recommendation methods to demonstrate the benefits of integrating collaboration-network learning and workload-aware ranking.

4 HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Component diagram

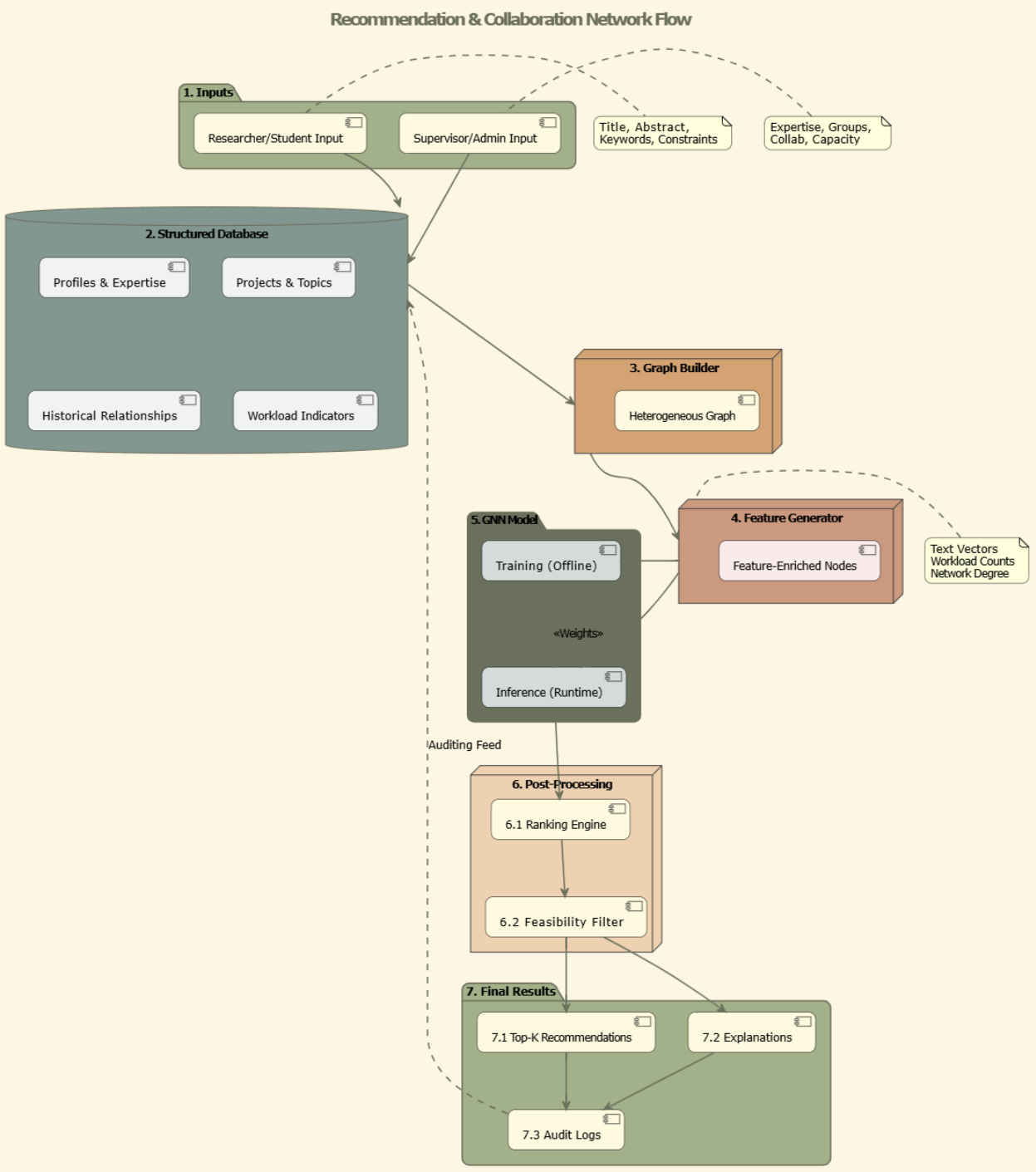


Figure 1: component diagram

4.2 Overview of the component

The **Recommendation & Collaboration Network** component supports researchers by recommending suitable supervisors (and optionally collaborators) for a submitted research idea. The component is built as a practical pipeline: it first collects and maintains structured supervisor/researcher profiles and research-idea metadata, then converts this information into machine-usable representations. Next, it builds a collaboration network that captures relationships such as research group membership, collaboration/co-authorship, co-supervision, and supervision history (where available). Using these signals, the component generates a ranked Top-K list of supervisors that are both **relevant** to the research idea and **feasible**, by incorporating workload/availability constraints in the final ranking. The key internal parts of this component are:

(i) profile & metadata storage,
(ii) (ii) representation generation,
(iii) (iii) graph/network construction,
(iv) (iv) graph/GNN scoring and ranking, and
(v) (v) feasibility filtering with explanation output.

4.3 Data flow

The Recommendation & Collaboration Network component supports researchers by recommending suitable supervisors (and optionally collaborators) for a submitted research idea. The component is built as a practical pipeline: it first collects and maintains structured supervisor/researcher profiles and research-idea metadata, then converts this information into machine-usable representations. Next, it builds a collaboration network that captures relationships such as research group membership, collaboration/co-authorship, co-supervision, and supervision history (where available). Using these signals, the component generates a ranked Top-K list of supervisors that are both **relevant** to the research idea and feasible, by incorporating workload/availability constraints in the final ranking.

4.4 Scalability and security considerations

For scalability, the design is data-driven: new supervisors, new expertise areas, and new research ideas are added through database updates, and the collaboration graph is rebuilt or incrementally updated without code changes. Using an inductive graph-learning style supports continuous growth, while periodic retraining (if required) can be scheduled rather than done on every update. For security, access to supervisor workload indicators and researcher-submitted content should be protected using role-based permissions, and recommendation outputs should be auditable through controlled logging. Privacy should be supported through data minimization, anonymization of identifiers during model training/evaluation, and secure handling of any sensitive profile information.

5 USER REQUIREMENTS

5.1 Stakeholder identification

The key stakeholders of the Recommendation & Collaboration Network component are:

- **Students / Early-stage Researchers:** provide research details and receive supervisor recommendations.
- **Academic Supervisors / Lecturers:** maintain profile information and may receive recommendation requests.
- **Department / Research Coordinators:** oversee supervisor allocation fairness, workload balance, and system governance.
- **University / Faculty Management:** benefit from improved supervision efficiency and transparency.

5.2 Functional requirements

FR1. The system shall allow a researcher to submit research details

FR2. The system shall store and manage supervisor profiles, including expertise areas and department/group information.
FR3. The system shall generate a ranked list of Top-K supervisors relevant to the submitted research details.
FR4. The system shall build and maintain a collaboration network using available relationships.

FR5. The system shall use the collaboration network to improve or refine supervisor recommendations.
FR6. The system shall consider supervisor workload and availability when generating the final ranked recommendations
FR7. The system shall provide a brief explanation for each recommended supervisor.
FR8. The system shall log recommendation results for evaluation and auditing purposes.

5.3 Non-functional requirements

- **Security & Privacy:** The system shall restrict access to sensitive profile and workload data and protect researcher-submitted research descriptions.

- **Performance:** The system shall return recommendations within an acceptable response time for typical usage.

- **Usability:** The input form and recommendation results shall be clear, simple, and understandable for students.

- **Reliability:** The system shall remain functional even if some profile fields or network relationships are missing .

- **Maintainability:** Supervisor profiles, expertise areas, and relationships shall be updateable without code changes.

- **Fairness:** The system should avoid repeatedly recommending the same small set of supervisors when multiple suitable options exist.

6 COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market is universities and higher education institutes that manage undergraduate final-year projects, postgraduate theses, and research supervision allocation. Secondary markets include private institutes, research centers, and EdTech platforms that support research supervision or academic collaboration. The main customer persona is a faculty/department research coordinator or academic administrator who is responsible for assigning supervisors, balancing workload, and improving research completion efficiency. A secondary persona is the student/researcher, who benefits directly from faster and clearer supervisor discovery.

6.2 Value Proposition

The solution provides a structured, data-driven way to recommend supervisors based on research-topic relevance, collaboration network signals**,** and workload feasibility. It reduces manual searching and repeated emailing, improves fairness by avoiding overload concentration on popular supervisors, and increases transparency by providing explainable ranked recommendations. This saves administrative time and improves the quality and feasibility of supervision matches.

6.3 Revenue Model

A practical revenue model is B2B SaaS licensing for institutions. Options include annual subscriptions per department/faculty or institution-wide licensing. An additional model is setup and customization fees for integrating with existing university, plus optional maintenance and retraining support packages.

6.4 Cost Estimation

Key cost areas include: initial development (engineering time), hosting (database + compute), periodic model training (if required), system maintenance, and customer onboarding/support. For a university-scale deployment, costs can be kept moderate using open-source tools and cloud infrastructure; the main ongoing costs are hosting and support/maintenance.

6.5 Pricing Strategy

Pricing can follow a tiered model based on institution size and features:

1. **Basic:** topic-based recommendation + simple workload filtering
2. **Standard:** collaboration network integration + explainability + dashboards
3. **Premium:** full GNN-based learning, advanced fairness controls, custom integrations, and analytics
   Pricing may be per year per department, with discounted institution-wide bundles.

6.6 Competitive Advantage

The key advantage is feasibility-aware recommendations: not only matching topic relevance, but also considering workload balance and collaboration structure to produce realistic recommendations. Many existing systems are either manual, text-only, or do not treat workload as a strong constraint. The pipeline enables scalability, easier updates when new supervisors register, and improved fairness compared to ad-hoc selection processes.

6.7 Intellectual Property Considerations

The main IP value lies in the system architecture**,** graph construction logic**,** ranking strategy**,** and workload-aware recommendation workflow, including any unique scoring formulas or training pipelines. While underlying libraries and algorithms are open-source, the implementation and dataset schema can be protected as proprietary. If institutional data is used, data ownership and privacy compliance must be clearly defined through agreements, and any trained models should avoid exposing sensitive personal information.

7. BUDGET AND JUSTIFICATION

This budget is proposed for developing, training, testing, and deploying the **Recommendation & Collaboration Network** component. All costs can be scaled down if university lab resources and free-tier services are used.

7.1 Hardware Costs

Cost: LKR 0 – LKR 200,000 (optional)
Justification: If a suitable GPU machine is already available (lab PC/personal laptop), no new hardware is required. If not, a mid-range GPU laptop/desktop or external GPU access may be needed to speed up GNN experimentation and training. This is optional because training can also be done on cloud GPU instances for short periods.

7.2 Software Licenses

Cost: LKR 0
Justification: The component can be implemented using open-source tools and frameworks.No paid licenses are required. If the institution requires specific proprietary tools, those would be covered by institutional licenses.

7.3 Cloud Services

Cost: LKR 20,000 – LKR 120,000
Justification: Cloud resources may be used for:

- hosting the database and API for testing/demos,
- running scheduled retraining jobs,
- short-term GPU compute for model training.
  Costs remain low if cloud is used only during training windows and free/educational credits are applied.

7.4 Human Effort Cost

Cost: Not billed (student research effort)
Justification: Development, experimentation, documentation, and evaluation will be completed as part of the one-year research project. If external support is required, additional staff hours may be needed, but this is usually covered by the department/university.

7.5 Summary Justification

Overall, the budget is kept minimal by using open-source software and limiting cloud usage to essential training and deployment tasks. The only potentially significant cost is GPU access justified by the need to efficiently train and validate the graph-based recommendation model.

8. WORK BREAKDOWN STRUCTURE (WBS)

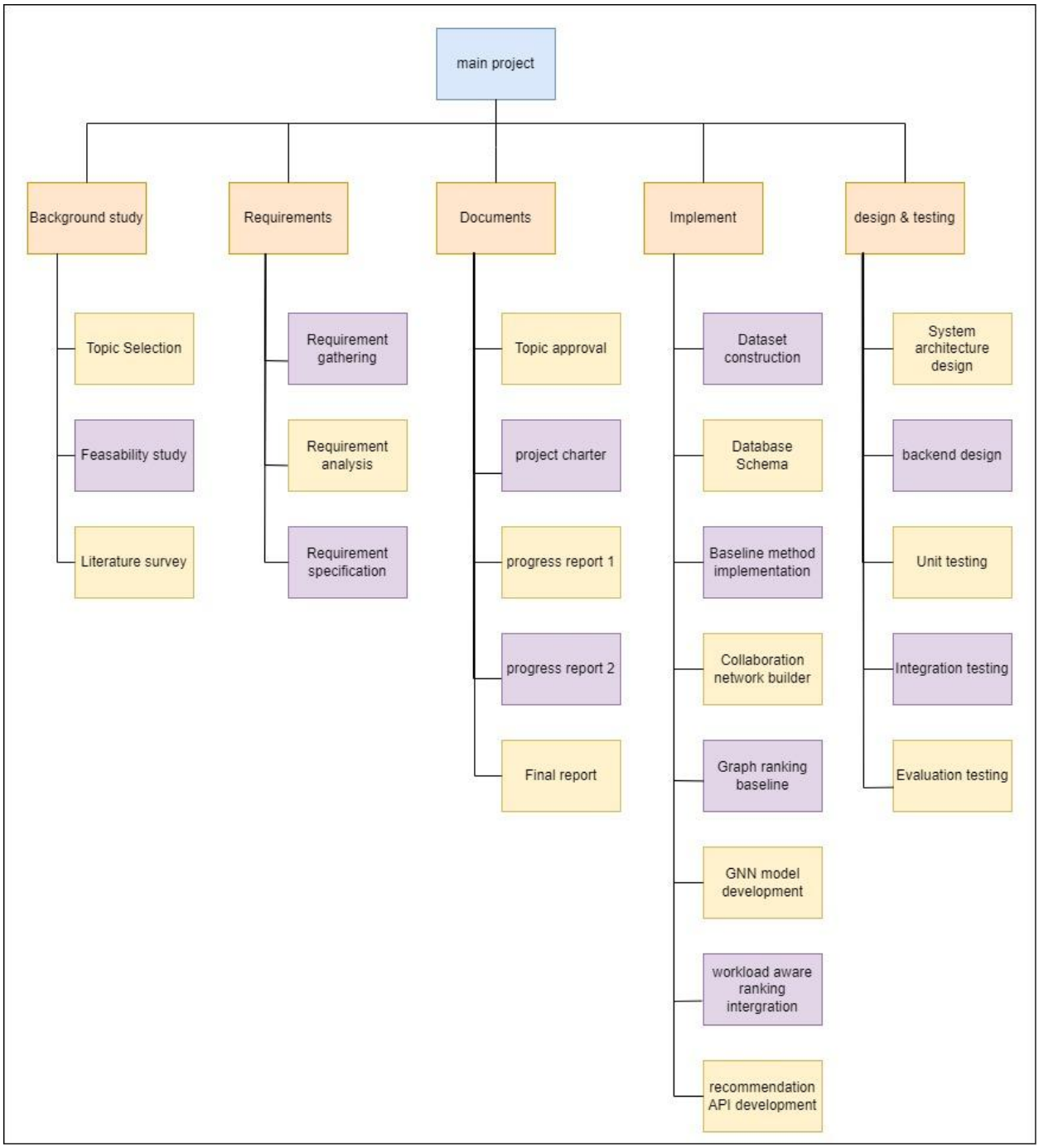


Figure 2: work breakdown structure

9. GANTT CHART

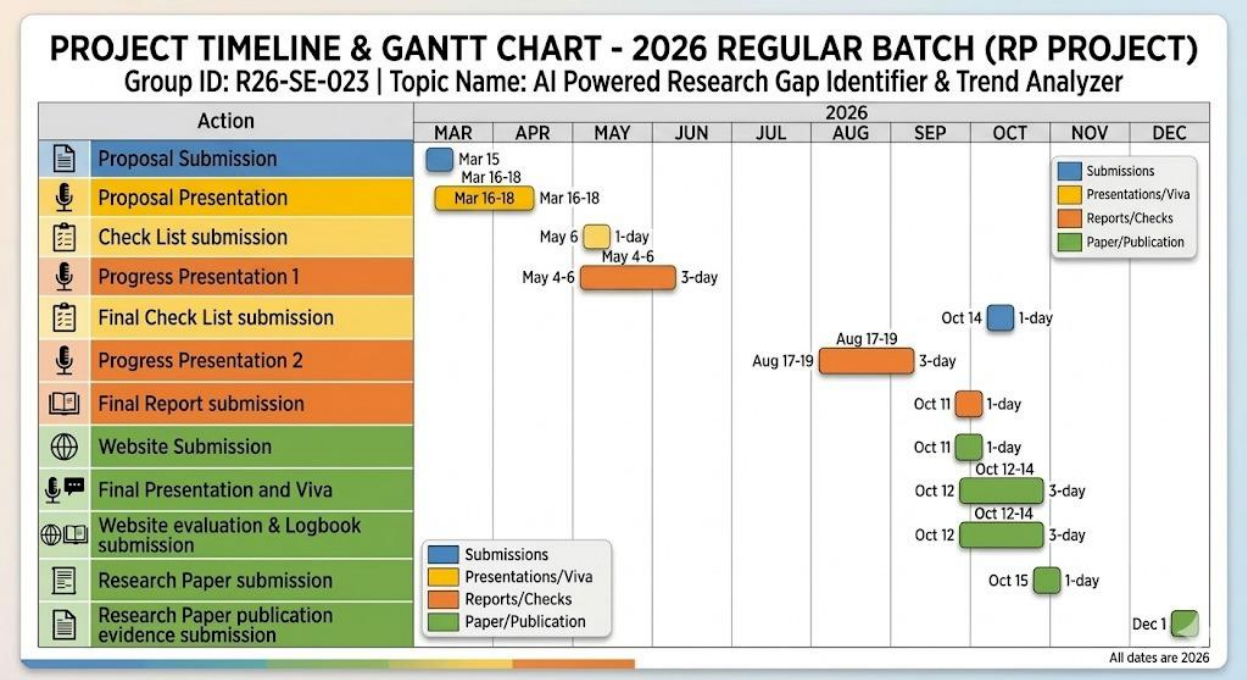


Figure 3: gantt chart

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11. APPENDICES

11.1 Preliminary UI mockups

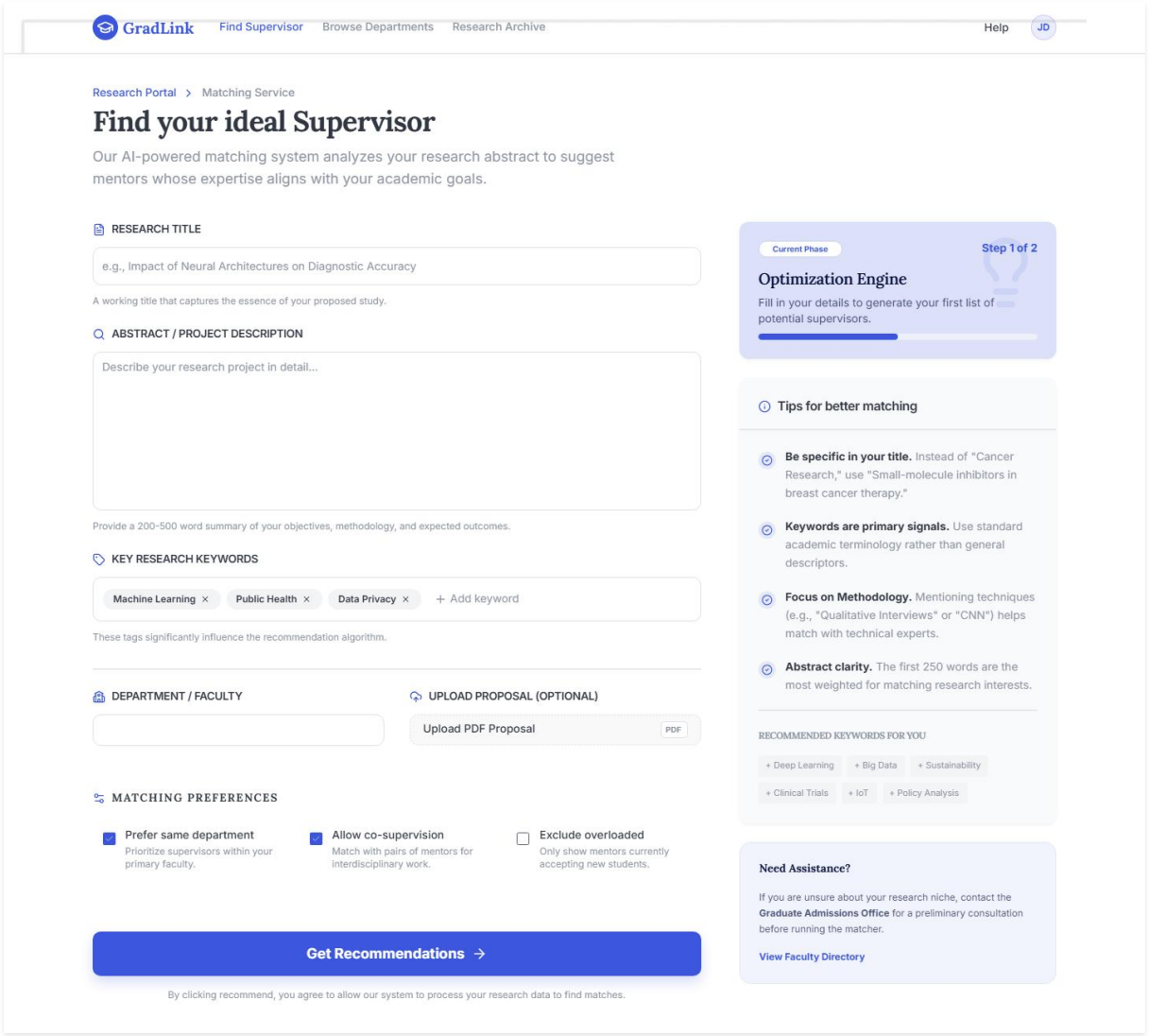


Figure 4: Research Details UI

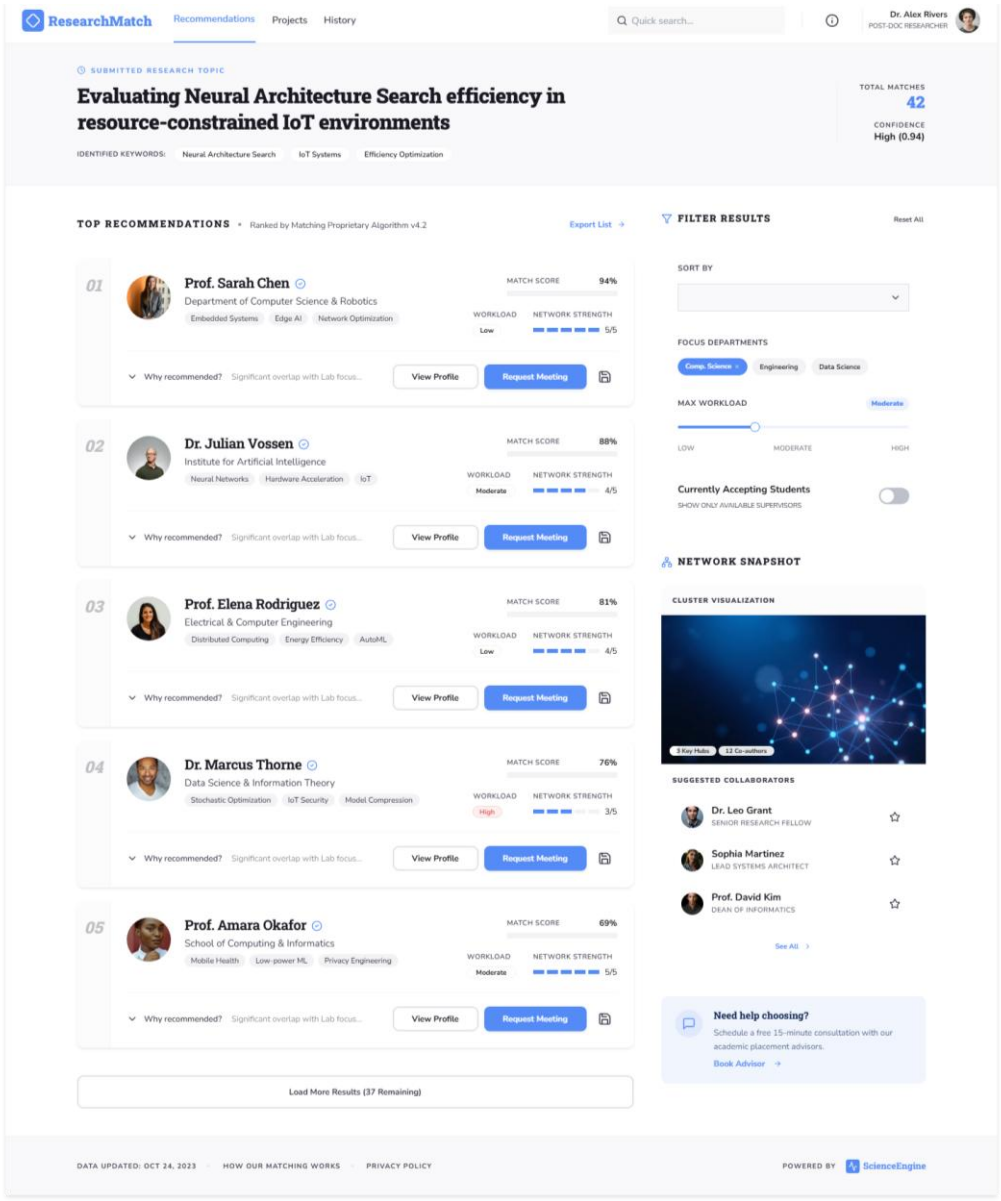


Figure 5:Supervisor recommendation UI

11.2 Risk analysis

Risk Description	Mitigation / Control
Insufficient data quality in supervisor profiles	Enforce minimum profile fields, allow periodic profile updates, auto-extract keywords from publications, and use fallback matching when fields are missing.
Limited historical supervision links for training the GNN	Start with strong baselines, and schedule retraining when more data accumulates.
Sparse or incomplete collaboration network	Use multiple relationship sources, add manual collaboration entry via admin UI, and apply graph ranking even with partial links.
Unreliable workload measurement	Use a clear workload proxy validate with supervisor/admin feedback, and treat GNN workload as improvement over proxy.
Bias / unfair recommendations	Add workload penalties/thresholds, add diversity constraints , monitor recommendation distribution, and report fairness metrics.
Privacy and ethical risks from using sensitive profile/workload or student topic data	Apply role-based access control, anonymize IDs for training/evaluation, minimize stored personal data, log access, and follow institutional approval procedures.
Scalability/performance issues	Use incremental graph updates, cache computed features, run training offline, and keep inference lightweight with Top-N shortlist retrieval.
Integration risks	Define clear API contracts early, version endpoints, use JSON schemas, and run integration testing with other modules.
Model maintenance risk	Schedule periodic retraining, monitor performance over time, and keep baseline ranking available as a fallback.
User trust issues	Provide short explanations, show workload/availability badges, and include transparent scoring factors at a high level.

Table 4: Risk Analysis